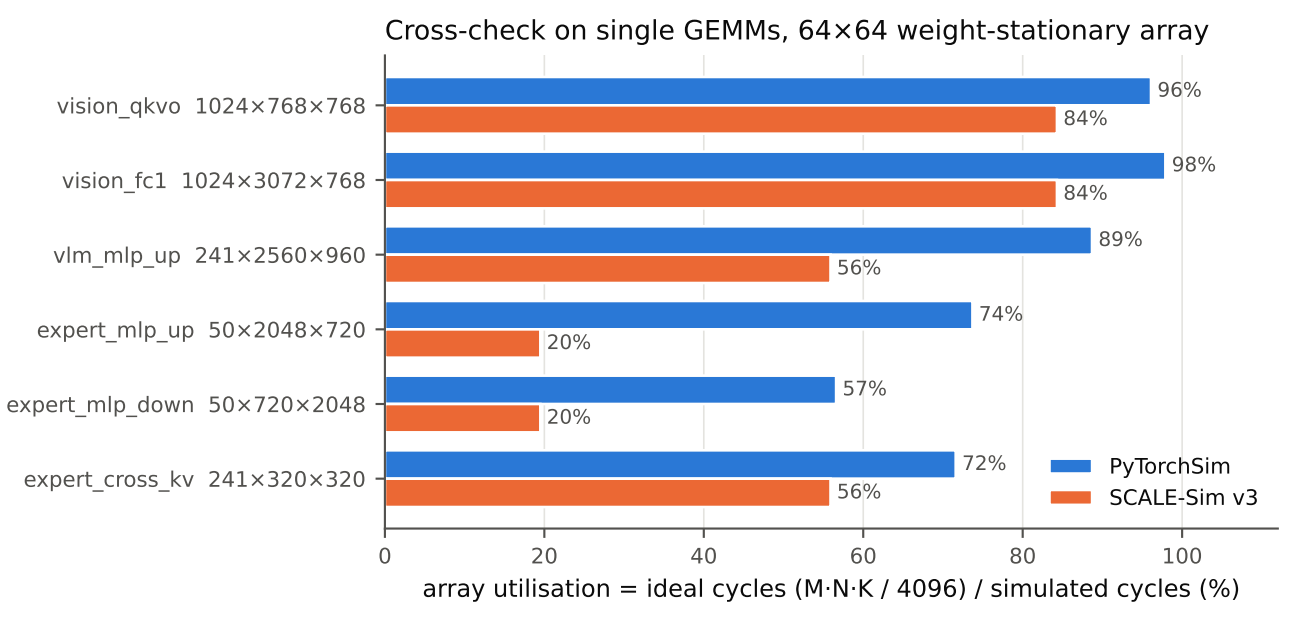
Cross-check on single GEMMs, 64×64 weight-stationary array
vision_qkvo 1024×768×768
vision_fc1 1024×3072×768
vlm_mlp_up 241×2560×960
expert_mlp_up 50×2048×720
expert_mlp_down 50×720×2048
expert_cross_kv 241×320×320
96%
84%
98%
84%
89%
56%
74%
20%
57%
20%
72%
56%
PyTorchSim
SCALE-Sim v3
0
20
40
60
80
100
array utilisation = ideal cycles (M·N·K / 4096) / simulated cycles (%)